Muhammad Asif Umar

AI Engineer | Software Developer | AI Solutions Engineer

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Portfolio | GitHub | LinkedIn

Professional Summary

Computer Science graduate specializing in Artificial Intelligence, with hands-on project experience spanning AI/ML, Generative AI, computer vision, Python automation, and full-stack development. Built web-based business applications, including e-commerce platforms and a POS system, alongside AI-driven solutions for emergency response, university security, parking management, and attendance monitoring. Experienced in data preprocessing, model training and evaluation, speech recognition, and integrating software components to address practical problems. Research-oriented with two submitted manuscript papers and selection for project presentation at INSC 2026. Seeking opportunities in AI engineering, software development, or AI solutions engineering.

Technical Skills

Programming: Python, C, C++, PHP, JavaScript, SQL, HTML, CSS

AI, ML and Deep Learning: Machine Learning, Deep Learning, Classification, Regression, Data Preprocessing, Feature Engineering, Model Training, Model Evaluation, TensorFlow, Keras, PyTorch, Scikit-learn

Generative AI and NLP: LLM Concepts, Prompt Engineering, AI Chatbots, AI Agents, Speech Recognition, Text Prediction, Grammar Correction, Natural Language Processing, AI API Integration

Computer Vision: OpenCV, Image Classification, Object Detection, Facial Recognition, Number-Plate Recognition, Image and Video Analysis

Full-Stack Development: Frontend and Backend Development, E-commerce Applications, Admin Dashboards, POS Systems, REST API Fundamentals

Databases and Data: MySQL, PostgreSQL, SQL, Pandas, NumPy

Security and Automation: Multi-Factor Authentication (MFA), Python Automation, GUI Development, Application Testing, Debugging

IoT and Connected Systems: GPS-Based Tracking, Emergency Resource Monitoring, Camera-Based Monitoring, Connected Device Concepts

Tools: Git, GitHub, Linux, VS Code, Jupyter Notebook, Google Colab, Docker Fundamentals

Professional Experience

Sukrat.ai – AI Product Development / Engineering *Internship / Product Development*

- Contributed to AI-powered product development, backend-related tasks, and improvements to application functionality.
- Investigated software defects, tested fixes, and evaluated application behavior to support product reliability.
- Supported AI agent workflows, adversarial testing, and edge-case validation to identify potential weaknesses in system behavior.

Software Development Projects

E-commerce Platform – Solarco
Full-Stack Development

- Developed an e-commerce platform for a solar-related business, covering the customer-facing storefront and administrative interface.
- Worked on application functionality, frontend/backend integration, product management workflows, and business requirements.

Multi-Shop E-commerce Platform
Full-Stack Development

- Built customer-facing and administrative interfaces for a multi-shop e-commerce application.
- Worked on storefront functionality, shop and product management, and administrative workflows.

Point of Sale (POS) Shop Application
Business Software Development

- Developed a shop-oriented POS application to support digitalization of retail operations and transaction workflows.

AI, Machine Learning and Generative AI Projects

AI-Driven Rescue 1122 Emergency Response System
Final Year Project | AI, Audio Classification, IoT Concepts

- Designed an AI-assisted emergency response workflow for call handling, incident classification, emergency record

creation, and dispatch coordination.
- Worked on audio dataset preparation and evaluation of SVM, CNN, and CNN–BiLSTM approaches for audio classification.
- Recorded experimental accuracies of 84.47% for SVM and 84.64% for CNN on the evaluated dataset.
- Designed an ambulance-tracking approach using GPS and connected-device concepts, with a focus on location reporting where mobile internet connectivity is limited.
- Explored citizen caller-location identification, emergency operator assistance, and response-team feedback as parts of the proposed emergency workflow.

Weed and Tobacco Detection Model
Machine Learning / Computer Vision

- Prepared data and trained a model for weed and tobacco detection.
- Worked on model training and evaluation for an image-based detection task.

Diabetes Prediction Using Machine Learning
Predictive Analytics

- Developed a classification workflow to predict diabetes risk from structured data.
- Applied data preprocessing, feature preparation, model training, and predictive performance evaluation.

Pashto Speech Recognition Module
Python | OpenAI Whisper

- Configured Whisper-based transcription to convert Pashto audio into text.
- Prepared the speech-to-text workflow for integration with downstream NLP and emergency-information extraction.

Python Automation and GUI Applications
Python Programming

- Developed Python utilities and GUI-based applications to automate tasks and provide interactive user interfaces.
- Explored next-word prediction and grammar correction workflows using language-processing techniques.

Computer Vision, IoT and Real-World Systems

University Smart Parking System
Python, Computer Vision, Number-Plate Recognition

- Worked on a web-based parking-management concept incorporating Python-based number-plate recognition to support vehicle identification and parking workflows.

University Entry and Exit Monitoring System
Facial Recognition and Fingerprint-Based Verification

- Worked on an entry/exit system combining facial recognition and fingerprint-based identity verification.

University Camera Monitoring and Movement Analysis
Computer Vision, Video Monitoring

- Explored integration of university camera feeds to identify or highlight students and monitor movement of people across monitored areas.

Hostel Mess Attendance System
Attendance Management, Application Development

- Worked on a hostel mess attendance solution to support attendance recording and monitoring.

Additional Projects

- **Multi-Factor Authentication (MFA):** Developed a Python-based authentication project focused on adding an additional verification layer.
- **Facial Recognition and Fingerprint Projects:** Explored biometric verification for access-control workflows.
- **Python Automation and GUI Tools:** Built utilities with graphical interfaces for different programming and task-automation requirements.
- **Web Applications:** Developed web-based solutions using PHP, MySQL, JavaScript, HTML, and CSS.

Education

University of Engineering and Technology (UET), Mardan *2026*
BS in Computer Science, Specialization in Artificial Intelligence
CGPA: 3.45/4.00

Research and Academic Achievements

- Two research manuscript papers submitted for academic consideration.
- Selected for project presentation at the 51st International Nathiagali Summer College on Physics and Contemporary Needs (INSC 2026).

- Recipient of three university merit scholarships.
- Recognized on the Dean's Award List for academic performance.
- Awardee of the Prime Minister's Laptop Scheme.
- Ranked among the top 20 students in the SSC-level Peshawar Board examination.

Certifications and Professional Development

Completed multiple training courses and certifications covering Artificial Intelligence, Machine Learning, Deep Learning, and Generative AI.

Professional Strengths

Analytical problem-solving, debugging, experimentation, technical research, model evaluation, software development, documentation, teamwork, and continuous learning.